

Multiple Choice (10 marks)

1. The risk factors for type 2 diabetes mellitus include:
 - (a) family history
 - (b) being overweight
 - (c) high intake of dietary fat
 - (d) All of the options listed are correct
2. What are histones?
 - (a) Lipids
 - (b) Carbohydrates
 - (c) Nucleotides
 - (d) Proteins
3. Since the 1996 World Food Summit, how has the number of food insecure people in the world changed?
 - (a) The Summit's goal of cutting the number of hungry people in half by 2015 was achieved.
 - (b) The number decreased, but not by nearly enough to meet the Summit's goal.
 - (c) The number increased slightly.
 - (d) Because of rising food prices, the number increased dramatically.
4. What is the main source of protein in the British diet?
 - (a) Fish and fish products
 - (b) Pulses
 - (c) Meat and meat products
 - (d) Milk and milk products
5. Which of the following will have the lowest glycaemic index?
 - (a) A baked apple
 - (b) A raw apple
 - (c) A raw potato
 - (d) Apple juice

True / False (10 marks)

Answer True or False

6. True or False: The doubly labelled water technique provides accurate assessments of all food intake, including low energy content foods.
7. True or False: Patients with Bulimia Nervosa typically have a body mass index (BMI) below 17.5 kg/m^2 .
8. True or False: Lactobacilli are the main bacteria that cause the onset of dental caries.
9. True or False: Demineralisation of tooth enamel occurs when the pH level is around 6.0.
10. True or False: Total urinary nitrogen excretion can accurately determine the daily amount of protein oxidized in the body.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the health implications of consuming vitamin C and other food supplements in excess. How can this affect overall health and what recommendations would you make for safe consumption?
12. Discuss the process of water transport between the intestinal lumen and the blood-stream during digestion, focusing on how it helps maintain osmolality and supports nutrient absorption.

End of Paper